Ex.No:03 Histogram of an images

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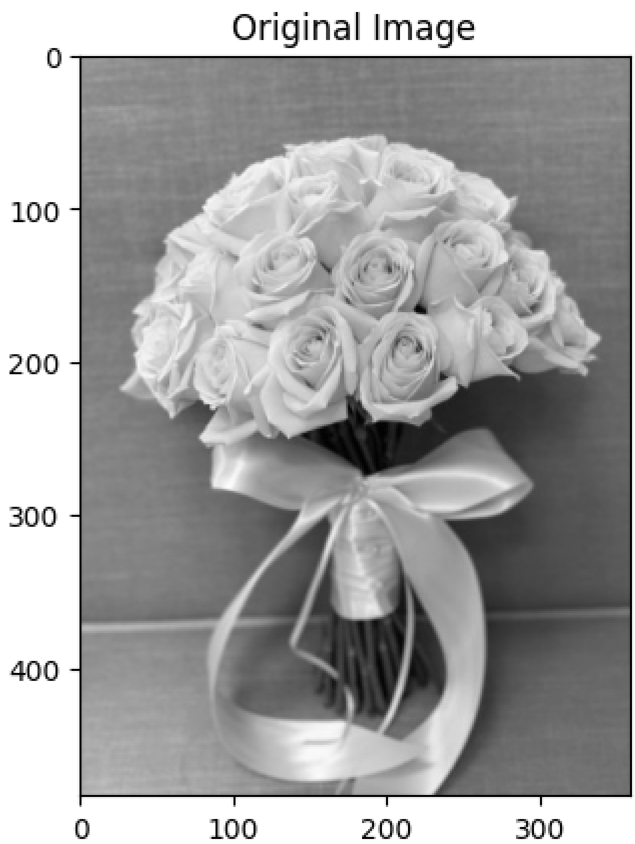
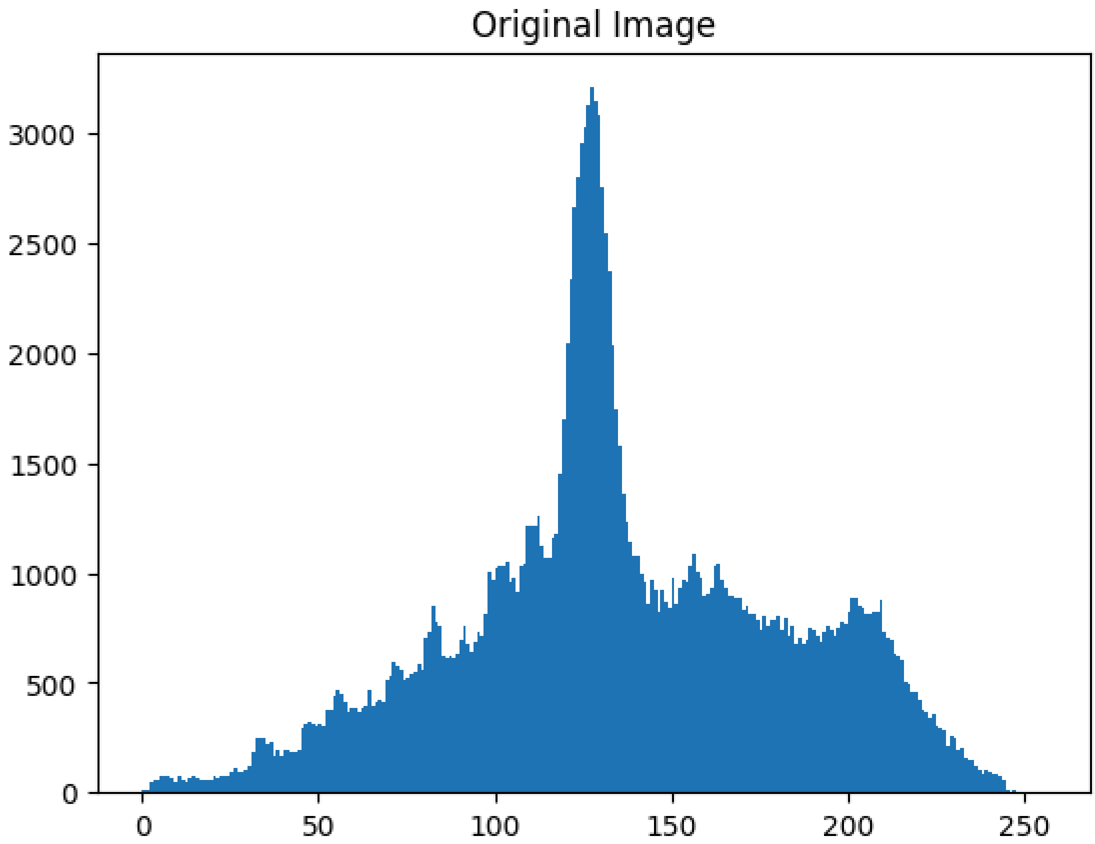
In [55]:
```
import cv2
import numpy as np
import matplotlib.pyplot as plt
img = cv2.imread('flower.jpg', cv2.IMREAD_GRAYSCALE)
plt.imshow(img, cmap='gray')
plt.title('Original Image')
plt.show()
```



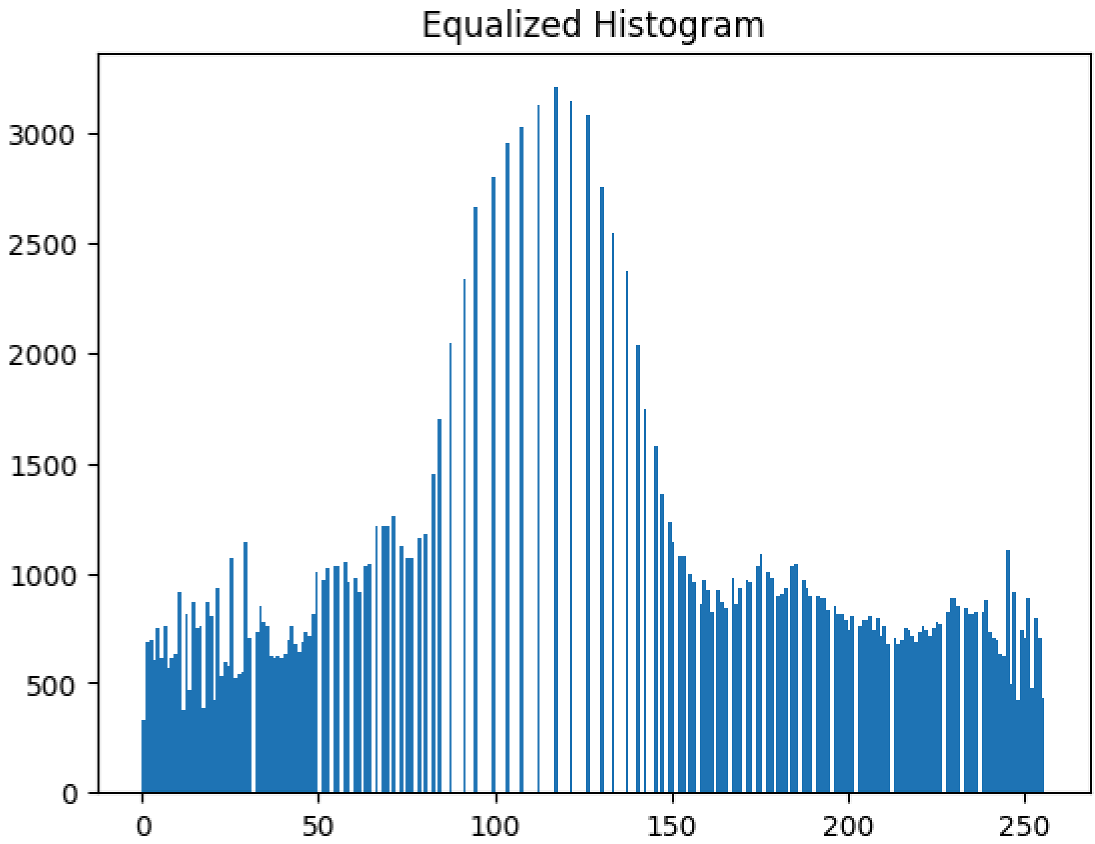
In [56]:
```
plt.hist(img.ravel(),256,range = [0, 256]);
plt.title('Original Image')
plt.show()
```



In [57]:
```
img_eq = cv2.equalizeHist(img)
```

In [58]:
```
plt.hist(img_eq.ravel(), 256, range = [0, 256])
plt.title('Equalized Histogram')
```

Out[58]:
```
Text(0.5, 1.0, 'Equalized Histogram')
```



In [59]:
```python
plt.imshow(img_eq, cmap='gray')
plt.title('Original Image')
plt.show()
```

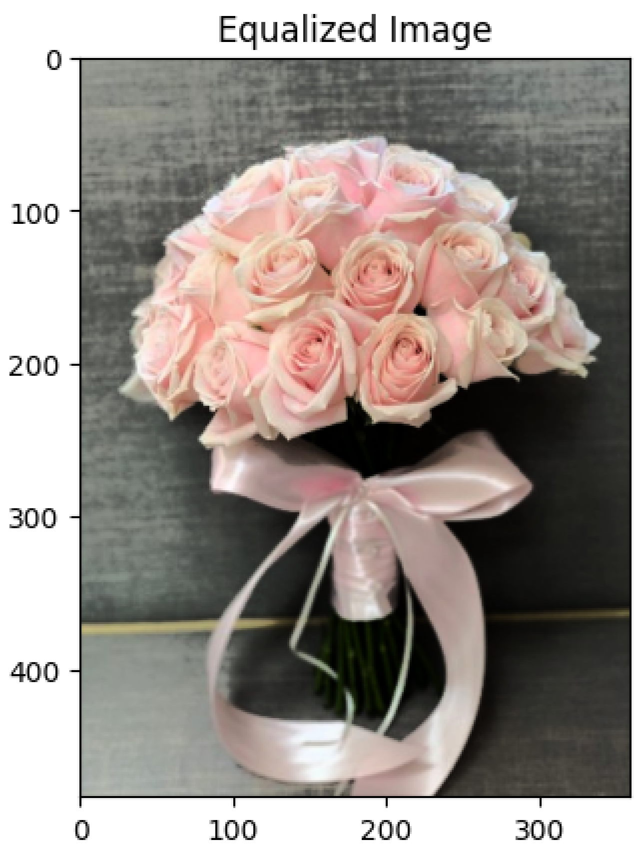
In [60]:
```python
img = cv2.imread('flower.jpg', cv2.IMREAD_COLOR)
```

In [61]:
```python
img_hsv = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)
```

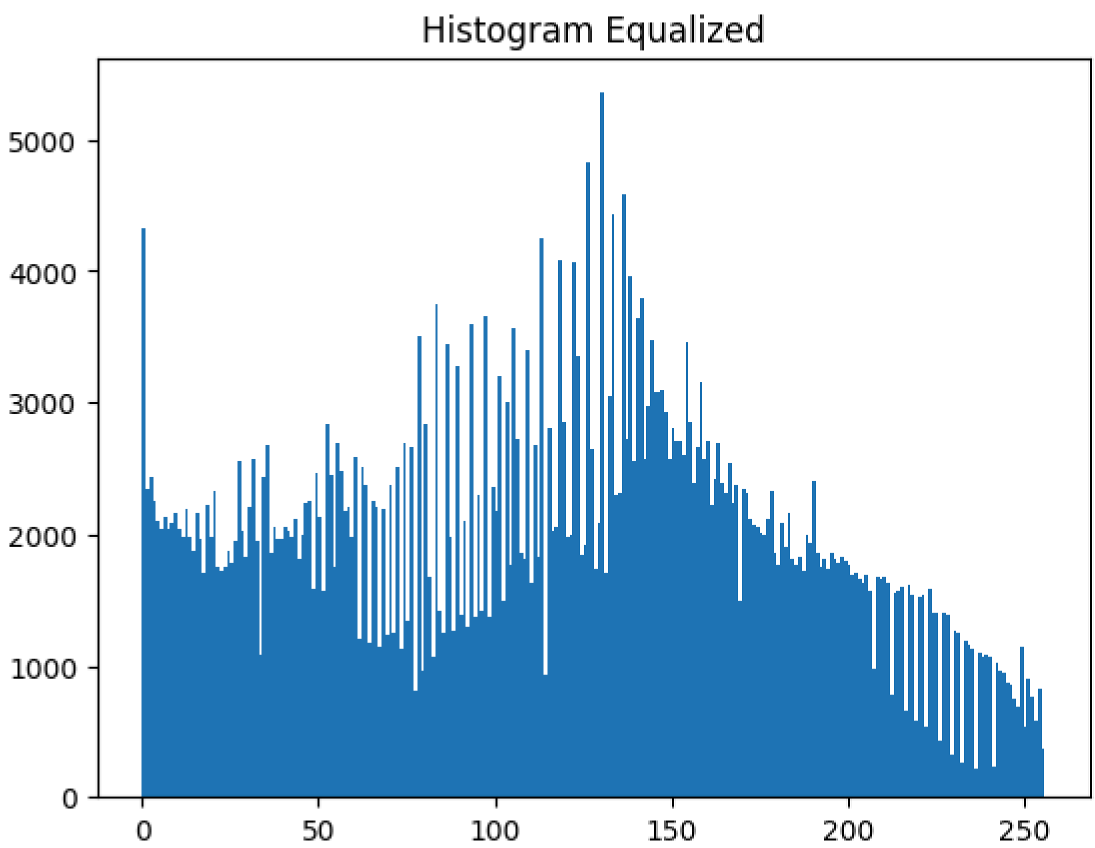
In [62]:
```python
img_hsv[:,:,2] = cv2.equalizeHist(img_hsv[:, :, 2])
```

In [63]:
```python
img_eq = cv2.cvtColor(img_hsv, cv2.COLOR_HSV2BGR)
```

In [64]:
```python
plt.imshow(img_eq[:,:,::-1])
plt.title('Equalized Image')
plt.show()
```

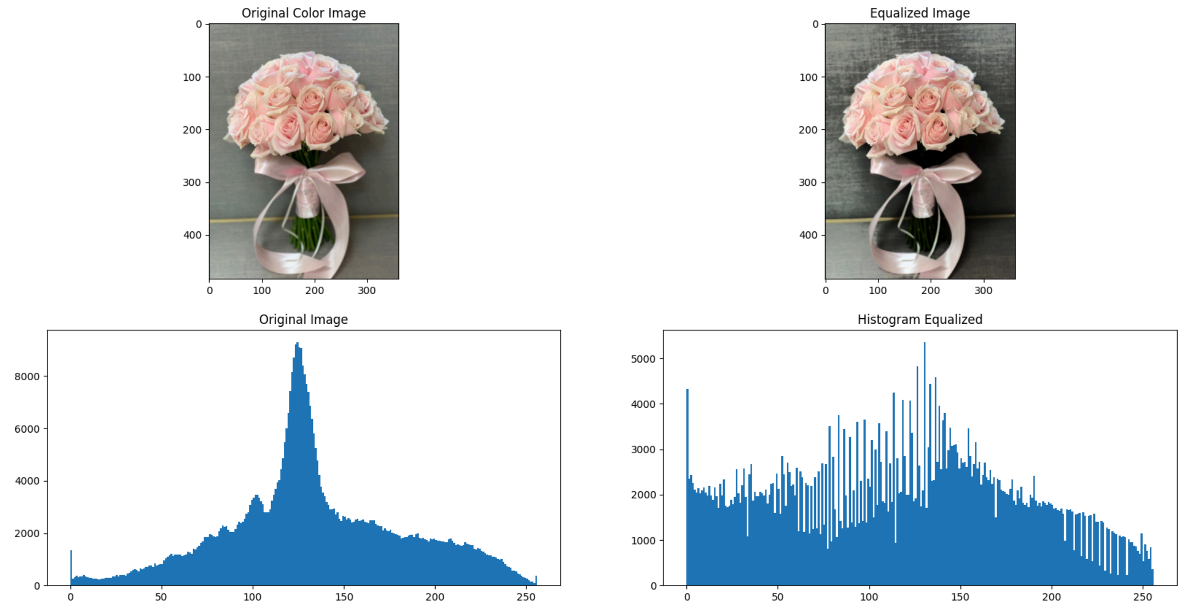


In [65]:
```
plt.hist(img_eq.ravel(),256,range = [0, 256])
plt.title('Histogram Equalized')
plt.show()
```



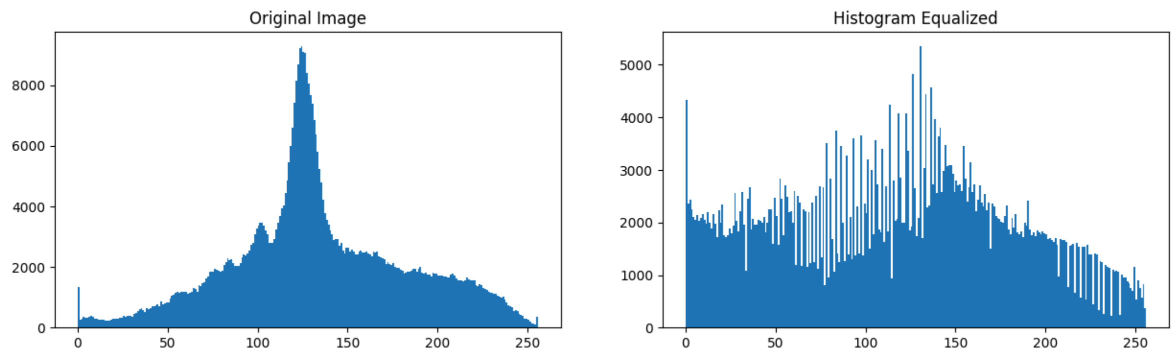
In [66]:
```
plt.figure(figsize = (20,10))
plt.subplot(221)
plt.imshow(img[:, :, ::-1])
```

```
plt.title('Original Color Image')
plt.subplot(222)
plt.imshow(img_eq[:, :, ::-1])
plt.title('Equalized Image')
plt.subplot(223)
plt.hist(img.ravel(),256,range = [0, 256])
plt.title('Original Image')
plt.subplot(224)
plt.hist(img_eq.ravel(),256,range = [0, 256])
plt.title('Histogram Equalized');plt.show()
```



In [67]:
```
plt.figure(figsize = [15,4])
plt.subplot(121)
plt.hist(img.ravel(),256,range = [0, 256])
plt.title('Original Image')
plt.subplot(122)
plt.hist(img_eq.ravel(),256,range = [0, 256])
plt.title('Histogram Equalized')
```

Out[67]:
```
Text(0.5, 1.0, 'Histogram Equalized')
```



In [ ]: